

Economics

AN AMELIA WEAVER GUIDE

Lunch 'aint
free, but this
is

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1 Part One: The ‘Rational’ Consumer (get it?)

Opportunity Cost lies at the heart of economics. The opportunity cost of doing something is what is given up. “There’s no such thing as a free lunch.”

For example

1. If your friend offers you the choice between eating a croissant and an eclair, by eating the croissant you forego eating the eclair.
2. If you can choose between reading a book and watching a movie and you choose to read, your opportunity cost is your viewing of the movie.
3. If you can choose to watch a movie given that you can read a book you own for an hour and a half or pay \$9.99 to watch a movie for an hour and a half, you forego the chance to spend the money elsewhere and reading time. *Opportunity Cost:* Spending \$9.99 elsewhere + the chance to read the book.

Note that there are two costs within the opportunity cost: the *explicit* and *implicit* cost. Explicit costs involve money and resources while implicit costs are more abstract. Paying \$9.99 for a movie is an explicit cost. Not getting the chance to read is an implicit cost.

Real Life Application: You want to watch a movie, so you go down to your local Blockbuster. You have the choice between renting the VHS for \$4.99 (five nights) or buying the VHS for \$9.99. You decide to rent. Your opportunity costs include the chance at spending \$4.99 on something else (explicit), a higher quality viewing experience due the degraded rental tape (implicit), and the ability to watch the movie as many times as you wanted after the rental period (implicit). You’re giving up the chance to spend money in other ways, higher quality, and, well, future opportunities.

Economics and Dungeons and Dragons are similar because there are fantasy elements in both, like the assumption that **people are rational**. One form of rationality is when it comes to **marginal decisions**. For example,

An alien invasion is taking place. In the mass panic, you grab your bike and start riding away from the chaos. Unfortunately, there are diminishing returns the further you ride your bike (the crowds are panicked). You know you can walk 5 km/h. In hindsight, when should you have hopped off your bike and began to walk?

Time	Speed (km/h)
11:00	22
12:00	18
13:00	15
14:00	9
15:00	4
15:00	2

It’s clear that you should have done so at 15:00. You could have covered 5 km instead of 4. This is the idea of the marginal decision.

Let's examine a more realistic scenario, involving work and pay. The **shadow government** has hired to you to work odd jobs around the compound. You can spend the hour feeding sharks or baking for the office. They pay you \$10 per shark you feed and a fixed rate of \$45 to bake for an hour. Here's your average shark record:

Time	Total Sharks Fed	Additional Sharks Fed	Profit
8:00	20	20	\$200
9:00	30	10	\$100
10:00	45	5	\$50
11:00	48	3	\$30
12:00	49	1	\$10

The sharks swarm for breakfast, but don't bite around lunch. It's clear that you should start baking at 11:00. If you just feed sharks, you make \$30 instead of \$45.

1.1 Macro Versus Micro

Microeconomics revolve around households and firms making decisions within a market. Macroeconomics study the overall economy, like its inflation rate, unemployment rate, and growth. For example,

Questions in Microeconomics include

1. What impact do minimum wage laws have on the labor market?
2. How do cake shortages affect the market for birthday parties?
3. What impacts individual savings decisions?
4. How do firms make production decisions?

Questions in Macroeconomics include

1. What causes economic recessions?
2. Are there policies that can help reduce economic recessions?
3. What makes economies grow?
4. How do heists of federal banks impact the economy?
5. How much should the government spend on public goods provision?

1.2 Positive Versus Normative

Positive statements are descriptive, confirmed, and testable statements:

1. The rate of unemployment is 4 percent.
2. A household swimming pool is 100 times more likely to lead to an accidental child death than a household firearm. (Read Freakonomics)
3. 3 percent of Americans ski and/or snowboard.

Normative statements are speculative arguments that aren't able to be confirmed or refuted, made by people named Tiffany who are too busy hanging out with their boyfriends to go skiing:

1. The rate of unemployment is too high.
2. The pool tax should be higher than the gun tax.
3. "Just go with Samuel and Jaya, you'll have an equally good time, and we can go another day."

Notice the parallel between the positive and normative statements. Statistics usually (well, they should usually) have backing. While an economist can make an argument for how the unemployment rate is too high, it's still up to be confirmed or refuted. An opinion, like the second point, is just the speaker stating what they think, not the reality. Similarly, the third statement (while potentially true) isn't concrete.

1.3 Examples

1. **Q:** Suppose you can either choose to go to the movies which cost \$7 for a 2 hour movie or you can babysit for your neighbor who pays you \$10 per hour. What is your total cost or opportunity cost of choosing to go to the movies? **A:** \$27
2. **Q:** Your work offers you the chance to work 2 hours of overtime this weekend at \$15 an hour. Instead, you choose to see a 2 hour movie. You pay \$8 for a ticket and \$5 for a snack. What is your opportunity cost of seeing the movie? **A:** \$43
3. **Q:** (T/F) Governments can never improve market outcomes. **A:** False, they so totally can
4. **Q:** Classify the question — 'What factors impact the US savings rate?' **A:** Macroeconomics
5. **Q:** Classify the question — 'What determines the quantity of shoes supplied in the market for shoes?' **A:** Microeconomics
6. **Q:** Classify the question — 'What's up with Black Friday?' **A:** Microeconomics
7. **Q:** Classify the statement — 'We should reduce the amount of federal taxes that individuals pay.' **A:** Normative
8. **Q:** Classify the statement — 'The Federal Reserve Bank voted to raise the target Federal Funds Rate in December 2015.' **A:** Positive
9. **Q:** Classify the statement — 'Greece must default on its debt if it is to recover economically from the severe depression it is currently in.' **A:** Normative
10. **Q:** Classify the statement — 'The U.S. needs to increase its defense spending to be prepared to protect against terrorist attacks.' **A:** Normative
11. **Q:** Classify the question — 'How would an increase in the taxes on gas impact the market for gas?' **A:** Microeconomics

2 Part Two: The Frontier

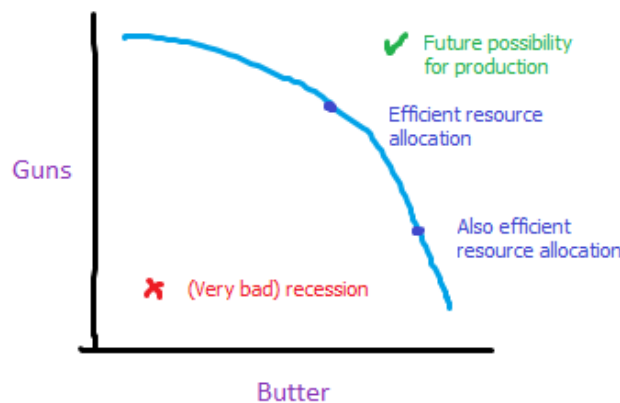
2.1 Society's Four Resource Categories

1. Labor: Human time spent producing goods and services.

2. Capital: Long-lasting, man-made tools that is factor into the production of other goods and services.
Human capital: The skills and training of the labor force.
Physical capital: Physical goods like machinery, equipment and factories.
3. Land: The physical space on which production takes place, as well as the useful materials (natural resources) found under it or on it, i.e. crude oil, iron, coal, soil, etc...
4. Entrepreneurship: The ability and willingness to combine the other resources into a productive enterprise.

2.2 Production Possibilities Frontier

The curve that defines the most efficient combination of production of two goods.



2.2.1 Assumptions

1. Fixed Resources: No changes in resources
2. Fixed Technology: No changes in technology
3. Productive Efficiency: When a society can not produce more of one good without reducing the production of another good
4. Full Employment: All available factors of production are being used to their maximum potential
5. Only 2 Goods: The society only has 2 goods available to produce

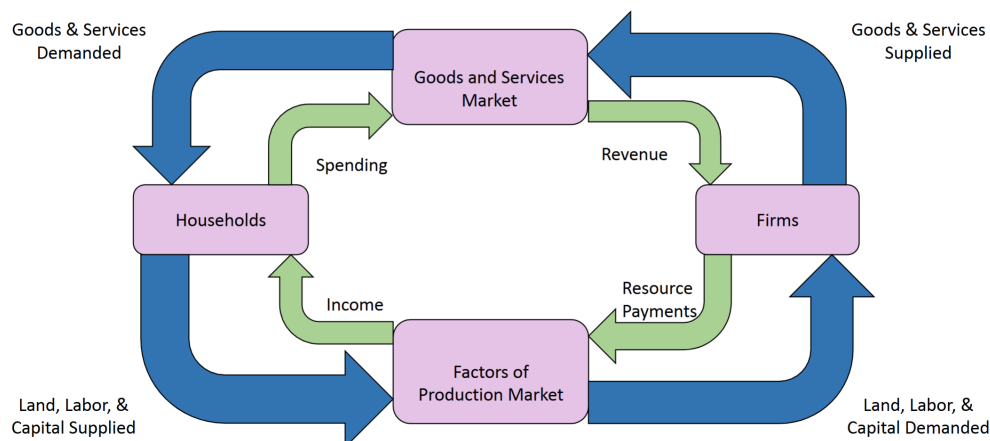
Recessions: A recession is represented as a point below the curve, not a shift in the curve.

Shifts: A shift inwards represents a loss of technology and/or available resources while a shift outwards is a gain in technology and/or available resources.

Cause of the Curve: The frontier is bowed outwards because allocating more and more resources to the production of a certain good renders diminishing returns. A linear curve indicates a constant opportunity cost, which means that the resources spent on both goods are interchangeable. For example, producing guns and butter require very few overlapping resources while producing cars and buses require various overlapping resources.

Opportunity Cost Eq.: Opportunity Cost of Producing Good A = $\frac{\text{Good B Diff}}{\text{Good A Diff}}$

2.3 Circular Flow Diagram



2.4 Examples

1. **Q:** Steel used in the construction of some goods is classified as what type of input? **A:** Physical Capital (steel is man-made)
2. **Q:** Attending College is considered what type of input in the production process? **A:** Human Capital
3. **Q:** (T/F) The PPF Curve can tell us about the causes of productive inefficiency. **A:** False
4. **Q:** (T/F) According to the PPF Model, the only 2 factors that influence economic growth are changes in technology and changes in land. **A:** False
5. **Q:** What concept(s) does the PPF Curve illustrate? **A:** Opportunity cost, growth, and scarcity
6. **Q:** (T/F) The PPF Curve provides an explanation for economic recessions. **A:** False
7. **Q:** Unrefined oil, from the ground, is an example of what type of factor of production (input) in the production process? **A:** Land (Natural Resources)

3 Part Three: To Trade Or Not To Trade

3.1 Specialization

Society is as advanced as it is due in part to specialization. Specialization allows for a society to

1. Develop Expertise: you don't need to churn your own butter (or faux butter if you're lactose intolerant), grow your own food, etc...
2. Minimize Downtime: switching between drastically different activities takes energy.
3. Gain Comparative Advantage: people become masters of a craft, and can trade that skill for money, goods, and/or services

3.2 Trade Advantage

Absolute Advantage is the ability to produce a good or service using fewer inputs than another producer uses. **Comparative Advantage** is the ability to produce a good or service at a lower opportunity cost than another producer.

Good	US	Brazil
Meat	3 meat units	4 meat units
Cake	1 cake unit	2 cake units

In this scenario, Brazil has absolute advantage over the US when it comes to both Meat and Cake production. It gets more interesting when solving for the opportunity cost of each good. Opportunity Cost Good 1 = $\frac{\text{Good 2 Count}}{\text{Good 1 Count}}$

Good	US	Brazil
Meat	$\frac{1}{3} = 0.33$ cake units	$\frac{2}{4} = 0.5$ cake units
Cake	$\frac{3}{1} = 3$ meat units	$\frac{4}{2} = 2$ meat units

These indicate that the US forgoes 0.33 cake units per meat unit they produce and 3 meat units per cake unit they produce. The same applies for Brazil's productions. In this case, Brazil has comparative advantage in meat production ($0.33 < 0.5$) while the United States has comparative advantage in cake production ($3 > 2$).

Using this information, each country can glean what they would accept as a trade price. Justified prices range from the opportunity cost of one good's creation to the other country's opportunity cost to create that good. For example, the two countries would be willing to trade anywhere from 2-3 meat units per cake unit. Let's be more specific. The truth is that the US is willing to trade 3 *or fewer* meat units per cake unit and Brazil is willing to trade 2 *or more* meat units per cake unit. If the US was spending more than 3 meat units on a cake, it would be more efficient for them to just make a cake unit instead. If Brazil was receiving fewer than 2 meat units per cake, they don't break even with their production cost for a cake unit.

We all know that the cake is a lie, however, so let's use a different example.

Good	Indonesia	Australia
Coconuts	3 hours	6 hours
Fishing Poles	9 hours	8 hours

Here we have Indonesia and Australia's time to produce one coconut and one fishing pole. Absolute advantage goes to the country that spends the least time making a good. In this case, Indonesia is more efficient at making coconuts ($3 < 6$) and Australia is more efficient at making fishing poles ($9 > 8$). Since the metrics are in hours, the equation is the inverse of the first one Opportunity Cost Good 1 = $\frac{\text{Good 1 Hours}}{\text{Good 2 Hours}}$.

Good	Indonesia	Australia
Coconuts	$\frac{3}{9} = 0.33$ poles	$\frac{6}{8} = 0.75$ poles
Fishing Poles	$\frac{9}{3} = 3$ coconuts	$\frac{8}{6} = 1.33$ coconuts

The process for determining comparative advantage here is the same, as the values are the opportunity cost for creating one of Good A. Indonesia is more efficient at growing coconuts ($0.33 \text{ ppc} < 0.75 \text{ ppc}$, ppc indicating poles given up per coconut). Australia is more efficient at making fishing poles ($3 \text{ cpp} > 1.33 \text{ cpp}$). The range of prices is determined the same as before.

3.3 Examples

- Premise:** Suppose that in one hour of work, Pat can produce either 9 bushels of wheat or 3 yards of cloth. In the same amount of time, Quinn can produce either 8 bushels of wheat or 4 yards of cloth.

Q: Who has absolute advantage in the production of each good? **A:** Pat has an absolute advantage in wheat, and Quinn has an absolute advantage in cloth.

Q: What is Quinn's opportunity cost of producing 1 yard of cloth? **A:** 2 bushels of wheat

Q: Who has comparative advantage in the production of each good? **A:** Pat has a comparative advantage in wheat, and Quinn has a comparative advantage in cloth.

Q: What would be the range of acceptable trade prices for 1 yard of cloth? **A:** 2-3 bushels of wheat
- Premise:** Suppose Morgan requires 2 hours to produce 1 computer and 4 hours to produce 1 printer. Riley requires 3 hours to produce 1 computer and 5 hours to produce 1 printer. Each worker has the same amount of labor hours available.

Q: Who has absolute advantage in the production of each good? **A:** Morgan rules in both, but she's humble about it.

Q: What is Riley's opportunity cost of producing 1 printer? **A:** 1.7 computers

Q: Who has comparative advantage in producing computers? **A:** Morgan

Q: What would be the range of acceptable trade prices for 1 printer? **A:** 1.7-2 computers

3. **Q:** Why is it a cake unit and not a cake? **A:** Standards keep the world going around. Engineering!
4. **Q:** (T/F) The market for sandals is an example of a market for factors of production. **A:** False
5. **Q:** (T/F) If a country has absolute advantage in the production of both goods, then there will be no trade. **A:** False. Diplomacy actually works out well for both parties.
6. **Q:** What are examples of things not gained from specialization? **A:** Absolute Advantage, Using Fewer Resources

4 Part Four: Karl Marx' Bane

4.1 Perfectly Competitive Markets

4.1.1 Assumptions

Upholding the economist's continuing faith in humanity, here are the bold assumptions about a perfectly competitive market:

1. **The Infinite:** The market consists of many buyers and sellers such that no individual can have an impact on price. Instead, they are price takers.
2. **What Glass Ceiling?:** There are no barriers to enter or exit the market, i.e. all firms can freely enter and exit the market.
3. **By The Book:** All products are homogeneous, i.e. there is no difference between products sold by different sellers. (Ex: Every Eggs Benedict is the same).
4. **Omniscience:** Both buyers and sellers have perfect information. All firms have access to the same production capabilities and resources. All buyers know all prices offered by all sellers.

4.2 Laws of Supply and Demand

4.2.1 The Law of Supply

The quantity supplied is the amount of a good that sellers are willing and able to sell at a given price.

The Law of Supply states that as the price of a good increases, the quantity supplied increases and vice versa. Sellers are more eager to sell an item at \$20 than at \$10.

4.2.2 The Law of Demand

The quantity demanded of a product is how much of a good consumers are willing to pay at a given price.

The Law of Demand states that as the price of a good decreases, the quantity demanded of that good increases and vice versa. More people are willing and able to pay \$10 for a sandwich than they are to pay \$20 for one.

4.3 Market Equilibrium

Market Equilibrium at the intersection between the supply curve and the demand curve. If the price of a good is below the market equilibrium, there's a shortage for that good. If the price of a good is above the market equilibrium, there's a surplus of that good. People like a good deal but won't be as likely to buy something overpriced.

During a shortage, suppliers will increase their prices to take advantage of the crazed crowds. This will lead to the market price drifting back to equilibrium.

During a surplus, suppliers will decrease their prices in the hope that people will purchase their product instead of a competitor's. This will lead to the market price drifting back to equilibrium.

4.4 Examples

1. **Q:** At a price of \$5 per baseball, people want to purchase 1000 baseballs a month from Target. Suppose Target raises the price to \$7 per baseball and people only want to purchase 750 baseballs a month. True or false: this is an example of the law of demand. **A:** True
2. **Q:** This principle defines the the Law of Demand. **A:** What is, uh, 'As the price of a good falls there is an increase in quantity demanded for that good and vice versa?'
3. **Q:** Is it true that a decrease in the price of cars would increase the demand for cars? **A:** No. It would increase the *quantity demanded*.
4. **Q:** At a price of \$10 a hamburger, In-N-Out wants to sell 1,000,000 hamburgers. Suppose at a price of \$8 a hamburger, In-N-Out wants to sell 2,050,000 hamburgers. This is an example of the Law of Supply. **A:** False. In-N-Out wants to sell more burgers at a higher price. The stockholders are shaking their heads.
5. **Q:** Should you get In-N-Out right now? **A:** Up to you
6. **Q:** The fact that an increase in price leads to an increase in supply is known as **A:** a question full of red herrings. If you're thinking of the Law of Supply, you want the words 'quantity supplied' to be in there.
7. **Q:** Suppose the current price of your favorite fishing bait is \$60 (premium bait), but the equilibrium price of the good is \$46. What is this market state called? **A:** A surplus. Why would you spend \$60 on bait that you can buy cheaper somewhere else?
8. **Q:** True or false: the Law of Supply states that there is a positive relationship between the price of a good and the quantity supplied of a good. **A:** True. Again, they wanna make money.
9. **Q:** A price set above the market equilibrium price causes quantity demanded to exceed quantity supplied causing a shortage which puts upward pressure on prices. **A:** Nope, it's a surplus. \$60 bait just isn't worth it.
10. **Q:** What are some assumptions of the perfectly competitive market? **A:** Sellers don't set the price of the product, all products are exactly the same, and all companies have the same production capabilities.
11. **Q:** Is it true that an increase in the price of cars leads to an increase in the supply of cars? **A:** No, you're looking for *quantity supplied*.

5 Part Five: Shifty Business

Market manipulation isn't the only shifty business in economics: certain events will shift the supply and demand curves.

5.1 Curve Shifts

The supply curve shifts right when the supply increases. Note how the supply and not *quantity supplied* increases. *As a result* of the supply increasing, the quantity supplied at each price will increase.

The demand curve shifts right when the demand increases. Again, note how the demand and not the *quantity demanded* increases. *As a result* of the demand increasing, the quantity demanded at each price will increase.

The supply curve shifts left when the supply decreases. Again, note how the supply and not *quantity supplied* decreases. *As a result* of the supply decreasing, the quantity supplied at each price will decrease.

The demand curve shifts left when the demand decreases. Again, note how the demand and not the *quantity demanded* decreases. *As a result* of the demand decreasing, the quantity demanded at each price will decrease.

5.2 Shifting the Supply Curve

Shifts in the supply curve can occur due to

1. **Input Prices**

An increase in input prices causes a decrease in supply and vice versa.

2. **Technology**

Advances in technology of a production of a good causes an increase in the supply of that good, and vice versa.

3. **Number of Sellers**

An increase in the number of firms and/or sellers in a market causes an increase in the supply of that good.

4. **Sellers Price Expectations**

An increase in the expected future price of a good decreases the current supply of that good and vice versa. Sellers want to hold off their stock until they can sell it at a higher price, or will want to sell as much of their stock as possible before its price drops.

5. **Weather/Natural Disasters**

Favorable weather conditions increase the supply of an affected good while disasters cause a decrease in the supply of an affected good.

5.3 Shifting the Demand Curve

Shifts in the demand curve occur due to

1. **Income**

There are two types of goods: normal and inferior. Whether a good is normal or inferior affects its demand after a change in income. The demand for normal goods increases when income increases. These are items like cars, houses, organic chicken meat, and household plants. Inversely, the demand for inferior goods decreases as income increases. These are items like cup ramen and apartments.

Normal goods' price elasticity of demand is elastic while inferior goods' price elasticity of demand is perfectly inelastic.

2. Prices of Related Goods

The relationship of the good to its related good affects its demand after the price change. There are two types of related goods: complements and substitutes.

Complement goods are consumed alongside the original good. These are goods like shampoo and conditioner, toothbrushes and toothpaste, and Christmas trees and ornaments. An increase in the price of a complement decreases the demand for the original good while a decrease in the price of a complement increases the demand for the original good. Since they are consumed together, given that a consumer will spend the same on both, they'll end up spending more to get the same quantity of the first and have to get less of the second. Inversely, if a consumer is spending the same amount of money, they can afford more of the original good since the related good is cheaper.

Substitute goods are consumed in lieu of the original good. These are goods like headphones and earbuds, Ghirardelli and Betty Crocker brownie mix, and apples and oranges. An increase in the price of a substitute increases the demand for the original good while a decrease in the price of a substitute decreases the demand for the original good. Consumers will be more willing to spend money on a cheaper product.

3. Number of Buyers

An increase in the number of buyers for a good will increase the demand for that good and vice versa.

4. Buyers' Price Expectations

If buyers expect a good's price to increase in the future, the demand for that good will increase in the present as those buyers won't want to spend more money on that product. This behavior ends up speeding up (or causing) the good's price to increase as the quantity demanded increases, following the Law of Demand and the market response to a shortage.

If buyers expect a good's price to fall in the future, the demand for that good will decrease in the present as those buyers will want to spend less money on that product. This behavior ends up speeding up (or causing) the good's price to decrease as the quantity demanded decreases, following the Law of Demand and the market response to a surplus.

5. Tastes

The more people prefer a good, the higher the demand for that good will be, and vice versa.

1. **Q:** Why is riding the bus considered an inferior good? **A:** Because Americans hate public transport. **Q:** Why? **A:** Because the public transport system is ignored by those with the power to improve it. **Q:** Why? **A:** Because Americans hate public transport.
2. **Q:** Suppose the price of Tropicana orange juice increases. What happens in the market for Simply Orange juice? **A:** The demand for Simply Orange juice increases.
3. **Q:** Suppose the price of coffee beans falls. Should we expect that the demand for coffee filters also decreases? **A:** No, we shouldn't.
4. **Q:** A change in the price of shoes could be represented as what type of shift in the market for shoes? **A:** A movement along the supply and demand curves.
5. **Q:** Hamburger patties and hamburger lettuce are an example of what type of goods? **A:** They're complements.
6. **Q:** Suppose that the median income in the U.S. rises. What would you expect to happen in the market for televisions? **A:** The demand for televisions would increase causing the price and the quantity of televisions supplied and demanded to increase.
7. **Q:** Suppose buyers expect the price of housing to fall next month. What would you expect to happen in the market for housing today? **A:** The demand for housing would decrease causing housing prices and the quantity of homes supplied and demanded to decrease.

8. **Q:** Suppose the price of hot dog buns increases. What do you expect to happen in the market for hot dogs?
A: The equilibrium price and the equilibrium quantity of hot dogs bought and sold both decrease.
9. **Q:** Suppose sellers expect the price of hot chocolate to increase next month. Also suppose that more people, some of which drink hot chocolate, move into the area. What happens in the market for hot chocolate today?
A: Price increases and quantity is indeterminate.
10. **Q:** Suppose a robot is invented that can assemble TVs in half the time it takes individuals to assemble a TV. Also suppose that a recession causes average incomes in the US to fall. What happens in the market for TVs?
A: Equilibrium price falls and quantity is indeterminate.

6 Part Six: Math Time

Elasticity measures how responsive a quantity demanded or quantity supplied of a good is to a change in one of its determinants.

6.1 Price Elasticity of Demand

The Price Elasticity of Demand is a measure of how much the quantity demanded of a good responds to a change in the price that good.

For any good, the price elasticity of demand for that good measures how willing and able customers are to buy less of a good as its price rises. Curves can be elastic, inelastic, or unit elastic.

Elastic quantities move proportionately more than the price, i.e. small changes in price cause large changes in quantity. $Elasticity > 1$

Inelastic quantities move proportionately less than the price, i.e. small changes in price cause even smaller changes in quantity. $Elasticity < 1$

Unit Elastic quantities move proportionately to the price, i.e. the price and quantity change by the same amount. $Elasticity = 1$

Perfectly Inelastic quantities and price are uncorrelated, i.e. the price has nothing to do with the quantity, i.e. the price of a good increases while the quantity remains the same. $Elasticity = 0$

Perfectly Elastic price is essentially fixed, i.e. a good with a price greater than the fixed price has 0 demand while anything with a price less than the fixed price has infinite demand. $Elasticity = \infty$

6.1.1 Factors

1. Availability of Close Substitutes: Goods have an elastic demand if it's easy to switch from one good to a substitute in response to a price increase.
2. Necessities vs Luxuries: Necessities have inelastic demands while luxuries have elastic demands. People need food, housing, and healthcare to live. People do not need subscriptions, name brand items, and other such luxuries to live.
3. Definition of Market: A market's specificity affects the its demand's elasticity. Broad markets tend towards inelasticity while narrow markets tend towards elasticity. There are no close substitutes for beverages (a broad market), while there are close substitutes for the coffee market.

4. Time Horizon: Over a shorter time horizon, goods' demand tends to be more inelastic while over a longer time horizon, goods' demand tends to be more elastic.

6.1.2 Calculation

Percentage Change Method

$$\text{Price Elasticity of Demand} = \left| \frac{\% \text{ change in quantity demanded}}{\% \text{ change in price}} \right| = \left| \frac{\frac{Q_2 - Q_1}{Q_1}}{\frac{P_2 - P_1}{P_1}} \right|$$

Midpoint Method

$$\text{Price Elasticity of Demand} = \left| \frac{\left(\frac{\text{change in quantity demanded}}{\text{midpoint of quantity demanded}} \right)}{\left(\frac{\text{change in price}}{\text{midpoint of prices}} \right)} \right| = \left| \frac{\frac{Q_2 - Q_1}{\frac{Q_2 + Q_1}{2}}}{\frac{P_2 - P_1}{\frac{P_2 + P_1}{2}}} \right|$$

$$\text{Total Revenue} = \text{Price} * \text{Quantity}$$

6.2 Income Elasticity of Demand

$$\text{Income Elasticity of Demand} = \frac{\% \text{ change in quantity demanded}}{\% \text{ change in income}} = \frac{\left(\frac{Q_2 - Q_1}{Q_1} \right)}{\left(\frac{\text{Income}_2 - \text{Income}_1}{\text{Income}_1} \right)}$$

6.2.1 Normal vs Inferior Goods

Normal goods have positive income elasticity for demand because quantity demanded and income move in the same direction. These elasticities vary widely, however, as necessities have small elasticities while luxuries have large ones.

Inferior goods have negative income elasticity for demand because quantity demanded and income move in opposite directions.

6.3 Cross-Price Elasticity of Demand

$$\text{Cross-Price Elasticity of Demand} = \frac{\% \text{ change in quantity demanded of good 1}}{\% \text{ change in price of good 2}} = \frac{\left(\frac{\text{good}_1 Q_2 - \text{good}_1 Q_1}{\text{good}_1 Q_1} \right)}{\left(\frac{\text{good}_2 P_2 - \text{good}_2 P_1}{\text{good}_2 P_1} \right)}$$

6.3.1 Substitutes vs Complements

Substitutes have positive cross-price elasticity of demand since the price of good 1 and the quantity demanded of good 2 move in the same direction.

Complements have negative cross-price elasticity of demand since the price of good 1 and the quantity demanded of good 2 move in opposite directions.

6.4 Price Elasticity of Supply

The Price Elasticity of Supply is a measure of how much the quantity supplied of a good responds to a change in the price that good.

For any good, the price elasticity of supply for that good measures how willing and able suppliers are to sell less of a good as its price rises.

6.4.1 Factors

1. Time Horizon

In the short run, supply is typically inelastic as firms cannot easily change their scale of production by building new factories or destroying existing ones and they can't easily enter a new market or exit their current market.

In the long run, supply is typically elastic for the reasons inverse of the ones listed above.

Note that perfectly inelastic supply reacts the same as perfectly elastic demand: the price may fluctuate without a change in quantity supplied. Perfectly elastic supply supplies an infinite quantity above the fixed price and zero below the fixed price.

6.4.2 Calculation

Percentage Change Method

$$\text{Price Elasticity of Supply} = \left| \frac{\% \text{ change in quantity supplied}}{\% \text{ change in price}} \right| = \left| \frac{\frac{Q_2 - Q_1}{Q_1}}{\frac{P_2 - P_1}{P_1}} \right|$$

Midpoint Method

$$\text{Price Elasticity of Supply} = \left| \frac{\left(\frac{\text{change in quantity supplied}}{\text{midpoint of quantity supplied}} \right)}{\left(\frac{\text{change in price}}{\text{midpoint of prices}} \right)} \right| = \left| \frac{\frac{Q_2 - Q_1}{\left(\frac{Q_2 + Q_1}{2} \right)}}{\frac{P_2 - P_1}{\left(\frac{P_2 + P_1}{2} \right)}} \right|$$

6.5 Examples

1. **Q:** You should expect an insulin shot used in the management of diabetes to have what type of price elasticity of demand? **A:** Inelastic. Also, insulin only costs \$2-\$4 to produce (here's a source). Bruh.

2. **Q:** Over the next 50 years, we should expect oil (used in the production of gas) to have what type of price elasticity of demand relative to the price elasticity of demand today? **A:** Elastic.
3. **Q:** Why do BMWs have an elastic demand curve while water, Tylenol, and natural gas have inelastic demand curves? **A:** Despite what the advertisements say, you don't need a BMW. You do, however, need freshwater to live.
4. **Q:** You work for a BMW dealership. Your boss tells you that he has decided to raise the price of all of the BMWs, because he wants to increase the dealership's total revenue. This will likely increase the dealership's total revenue. **A:** False. Again, people don't need BMWs and the dealership across the street (why are car dealerships always next to each other?) is selling a different brand for cheaper. Also, they're next to each other because of Hotelling's Law.
5. **Q:** Suppose you work in the apple business, a good with many close substitutes. You want to increase your overall total revenue. If you lower the price of your apples, will you make bank? **A:** Yes. People will flock to your amazing prices. You may also cause a price war.
6. **Q:** In the market for laptops, 2500 laptops are demanded at a price of \$1,500 per laptop. However, 6000 laptops are demanded at a price of \$800 per laptop. Using the midpoint method, calculate the price elasticity of laptops. **A:** 1.35
7. **Q:** Suppose at a price of \$1 there are 50 bananas demanded, and at a price \$2 there are 38 bananas demanded. What type of elasticity can you deduce from this data? (Use the percentage change method) **A:** Inelastic
8. **Q:** Total revenue did what after the price increase in the prior Practice Question? **A:** Increased.
9. **Q:** Does the price elasticity of demand tell us how responsive price is to a one unit change in quantity demanded? **A:** No.
10. **Q:** Lowering the price of a good will increase total revenue if the demand curve is relatively elastic. **A:** Yes. People love a good deal.
11. **Q:** In general, we should expect that the demand for cars will be relatively more inelastic than the demand for Hondas (an affordable car brand). **A:** Yes.
12. **Q:** What type of elasticity do epipens' demand curve have? **A:** Inelastic. It's a life saving device. Also, epipens cost less than twenty dollars to produce (here's a source). Brand name EpiPens cost around \$600. Bruh.
13. **Q:** Think: will the price elasticity of supply always be positive? **A:** Yes.
14. **Q:** The income elasticity of demand is a measure of how much the demand of a good responds to changes in a consumers income. **A:** False. It relates the quantity demanded to income.